

Katrina-Kay Pettitt, Ph.D.

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SKILLS

- **Technical:** Python, SQL, C++, JavaScript, HTML, CSS
- **Tools:** Git (GitLab & GitHub), AWS (S3, EC2, MSK, Cloudwatch), GCP, Jupyter, Airflow, Kafka, Superset, Kibana, Docker, Kubernetes, Django, Flask, Elasticsearch, ArgoCD, DBeaver, Jira
- **Areas of Expertise:** ETL, data pipelines, data scraping, data architecture, data storage and retrieval, data analysis, data mining, data wrangling, cloud computing, API development and integration, data operations and strategy, data visualization, software design, database management (PostgreSQL, SQLite)
- **Languages:** English (native), Japanese (fluent), French (conversational)

WORK EXPERIENCE

Maternity Leave | July 2024 — Present

Career break to focus on my pregnancy, baby, and family. Returning to the workforce in 2025.

Software Engineer | March 2024 — July 2024

Comply Advantage | UK | Remote

- Develop asynchronous REST API applications with Python, SQLite, and FastAPI to replace internal reliability on third-party software
- Write thorough unit and integration tests with pytest and unittest to ensure reliability and performance
- Optimize scalability of projects covered by the Data Ingestion Team by improving code efficiency
- Collaborate with other engineers on software design and implementation

Data Engineer | June 2021 — March 2024 [Acquired by Comply Advantage]

Golden Recursion | California, USA | Remote

- Create, develop, maintain, and improve high-performance data ingest processes such as data pipelines for arXiv's academic article database, the USPTO patent database, and the ClinicalTrials.gov database, resulting in >80% of database contributions in 2023.
- Perform one-off bulk data imports such as the Open Library and uniRank databases using packages such as lxml, requests, Selenium, and Playwright
- Improve data quality through large scale data wrangling, analyses, and cleansing in Python and SQL
- Develop data architecture and schema for current and future data needs
- Conduct exploratory data analyses and create data visualizations to craft effective data strategies
- Manage the company's data acquisition teams, which respond to client data ingestion requests

Early career work experience from 2017-2021 as a Postgraduate Research Fellow and PR Specialist with Hokkaido University in Sapporo, Japan involved working on government-funded projects, web and media development (including front-end work in JavaScript, CSS, and HTML), automating administrative workflows, managerial responsibilities, and interpretation and translation duties in English and Japanese.

EDUCATION

PhD Classics and Archaeology | 2017 | University of Exeter | Exeter, UK

Received a full-ride scholarship to research ancient archaeological assemblages in Roman Britain. Handled, retrieved, and analyzed large amounts of data. Field experience in local archaeological projects. Postgraduate Teaching Assistant and tutor for classes on ancient history, archaeology, and Latin.